

What properties of a teacher guide can produce support for mathematics teachers?

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This presentation focuses on teacher guides, which together with textbooks play important roles in the work of mathematics teachers. However, mathematics education research on teacher guides is not as developed as research on textbooks (Jukić Matić & Glasnović Gracin, 2021). A review of research on teacher guides in any subject found only 43 studies from 1990-2020 (Buch et al., 2022). Results in the review are primarily at more general levels, for example, about teachers' views on guides, whether guides are used by teachers and how often and when. Few studies make specific links between properties of guides and the work of teachers. Our study contributes by developing design principles for teacher guides, in relation to principles for teaching and learning.

The aim of our ongoing study is to produce a deeper understanding of the role that teacher guides can play in supporting mathematics teachers, specifically addressing the following research question: How do different properties of a teacher guide affect teachers' work when planning and evaluating mathematics lessons?

Framework

We rely on design research methodology, including a focus on design principles, which refers to empirically and/or theoretically grounded descriptions and arguments of how to achieve a goal (van den Akker, 2010). We focus on two types of design principles.

First, we use design principles concerning teaching and learning, based on previous research. These design principles describe how the interaction between different forms of expression, or modes, in mathematics teaching can support students' learning, based on aspects of cohesion (e.g., similarity, continuity) and tension (e.g., contrast, distance) between modes. Our main design principle state that to promote learning, there must be a balance between cohesion and tension in the teaching situation, and tension between modes must be sharp enough to be challenging for the students (Engebretsen, 2012).

Second, through empirical analyses, we intend to develop design principles for a teacher guide. The purpose of our teacher guide is to support teachers to work in line with the principles of teaching and learning. Our study aims to create design principles on how properties of the teacher guide can enable teachers to receive this support.

Method

This study is based on two projects: One project with four teachers in lower primary school, delimited to subtraction, and one project with three teachers, one each from lower primary, upper primary, and lower secondary school, focusing on the concept of mathematical equality. In each project, we first met with the teachers to present and discuss the design principles for teaching and learning. Then followed cycles of testing

and revising a teacher guide of about two pages, where teachers and researchers met before and after lessons to jointly plan and evaluate the teaching. We recorded all meetings, covering 17 lessons, where 7 versions of the teacher guide have been used.

Data analyses clarify how properties of the teacher guide seem to influence teachers, which is done in two ways. First, teachers' explicit descriptions of the impact of properties are included. Second, we analyse how teachers relate to and (do not) use different parts of the guide in their work with planning, implementing and evaluating lessons.

Tentative results from ongoing analyses

So far, we have located some links between properties of the guide and teachers' work. In the project about subtraction, a change was made to the guide that seemed to support the teachers in a more concrete way to reflect on aspects of cohesion and tension to create a contrast between different modes. We changed from an overall focus on *expected challenges for the students*, with respect to using different modes, to *contrasts between modes* that are present in the teaching material. In the project about equality, revisions of the guide continued at a more detailed level, in line with the change of the overall focus. For example, when teachers talked about *contrasts for students* when planning lessons, we changed formulations in the guide, to instead of asking teachers to reflect on how students need to handle a contrast between modes, we asked in what way a *contrast between modes* should exist in the teaching material.

Based on these preliminary results, a tentative design principle can be formulated: A teacher guide for mathematics teachers should focus on supporting the teacher to reflect on properties of the teaching material to be used, since this seems to make the support more concrete and useful. Continued analyses will scrutinize this design principle in relation to other data and develop more design principles for teacher guides.

References

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